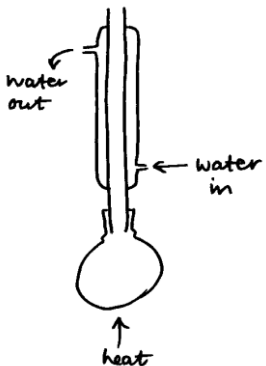


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		award (1) for either of following acidified (potassium) dichromate or $\text{Cr}_2\text{O}_7^{2-} / \text{H}^+$ acidified (potassium) manganate(VII) or $\text{MnO}_4^- / \text{H}^+$	1			1		1
		(ii)		accept formulae in any format e.g. $\text{CH}_2\text{OHCH}_2\text{OH} + 4[\text{O}] \rightarrow (\text{COOH})_2 + 2\text{H}_2\text{O}$ $(\text{CH}_2\text{OH})_2 + 4[\text{O}] \rightarrow (\text{COOH})_2 + 2\text{H}_2\text{O}$ $\text{C}_2\text{H}_6\text{O}_2 + 4[\text{O}] \rightarrow \text{C}_2\text{O}_4\text{H}_2 + 2\text{H}_2\text{O}$ correct formulae of reagents and products (1) balancing (1)		2		2		
		(iii)		 vertical condenser with flask being heated (1) water going through condenser in correct direction (1) award (1) only if condenser shown in distillation apparatus with water going from low to high	2			2		2